OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

13.13 Object CRG surface

In driving simulators, the driver should be able to feel both the general road surface and specific, visible features on the road. This includes manhole covers, cracks, and patches. However, these require different resolutions, and it is impractical to model the entire road surface with a fine grid.

To support high-resolution surface features on a low-resolution road, the `<surface>` element may be included within the `<object>` element.

Objects with a defined `<surface>` element influence the height of the road, including elevation, lateral profile, interpolated elevation grid, and lane height. The object's surface height is added to the road height. In the other direction, it is subtracted if the object's surface height is negative.

Because CRG data may only cover parts of a road's surface, it must be made sure that the elevation information derived from ASAM OpenDRIVE data can still be used outside of the valid CRG area. Because the reference line of the CRG file for an object is ignored, a CRG file may be referenced multiple times in different parts of the map. The local coordinate system of an object is rotated with the object. The object surface is thus also rotated.

Processing math: 100%

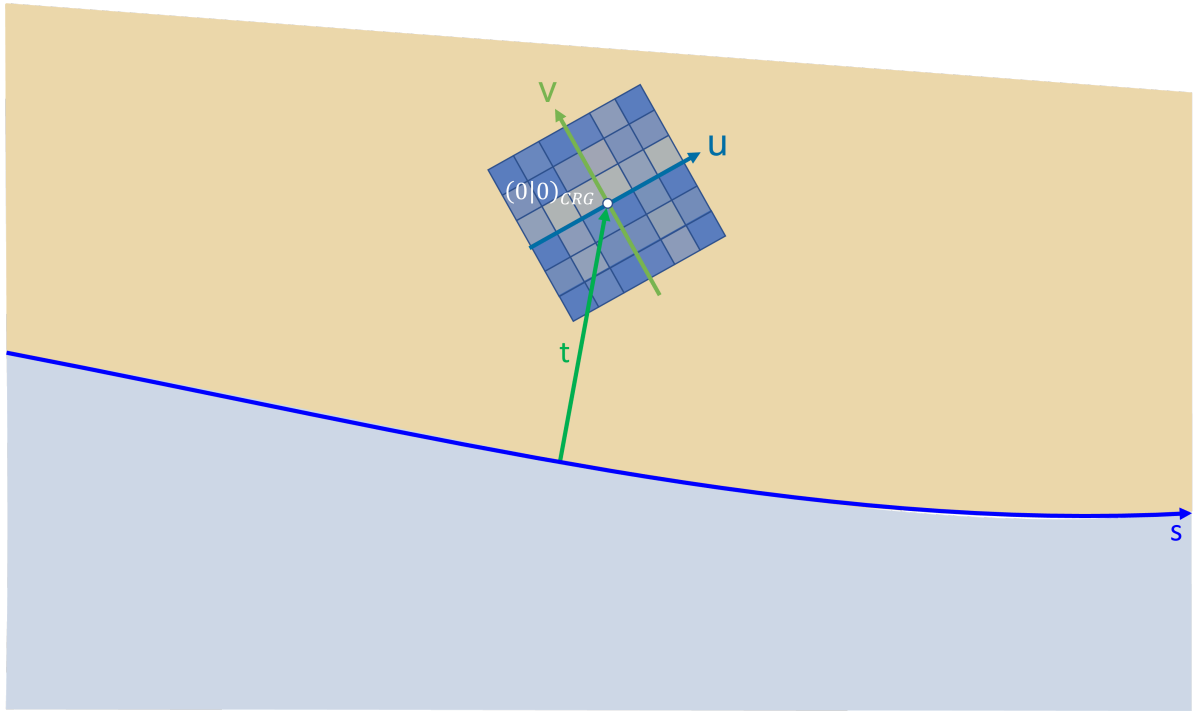


Figure 123. CRG for objects

Figure 123 shows how CRG for objects is applied. s/t is the road reference line coordinate system. u/v is the CRG coordinate system. The object's center point is defined as the origin of the CRG coordinate system, which results in positive and negative u/v values.

Elements in UML model

`<surface>` element

In ASAM OpenDRIVE, object surfaces are represented by the `<surface>` element within the `<object>` element.

```
UML class:  t_road_objects_object_surface
XML tag:    <surface> (Multiplicity: 0..1)
Introduced: 1.7.0
```

Processing math: 100% road surface elevation of an object.

`<CRG>` element

In ASAM OpenDRIVE, ASAM OpenCRG data of object surfaces is represented by the <CRG> element within the <surface> element.

```
UML class:  t_road_objects_object_surface_CRG
XML tag:    <CRG> (Multiplicity: 0..1)
Introduced: 1.7.0
```

Elevation data described in ASAM OpenCRG are represented by the `<CRG>` element within the `<surface>` element.

Table 106. Attributes of the <CRG> element

Name	Type	Use	Introduced	Description
`file`	string	required	1.7.0	Name of the file containing the CRG data.
`hideRoadSurfaceCRG`	t_bool	required	1.7.0	Determines if the object CRG hides the road surface CRG. Default is true.
`zScale`	double	required	1.7.0	z-scale factor for the surface description (default = 1.0).

XML example

```
<surface>
    <CRG file="manhole_cover.crg" hideRoadSurfaceCRG="true" zScale="1"></CRG>
</surface>
```

Calculation

If an ASAM OpenCRG file is referenced, the u/v coordinates of the object are used directly as u/v coordinates of the ASAM OpenCRG file. The reference line of the CRG file is unused.

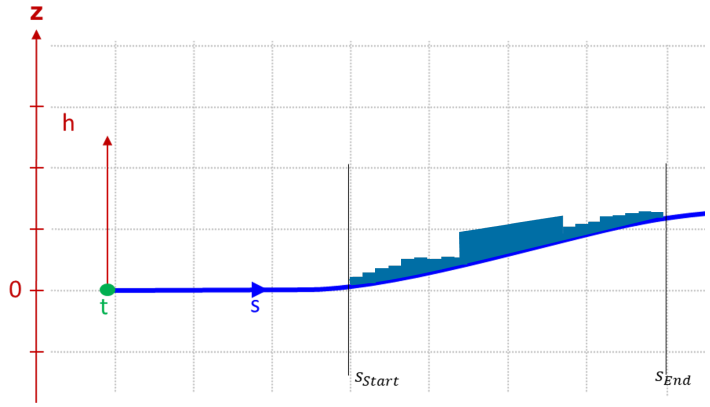
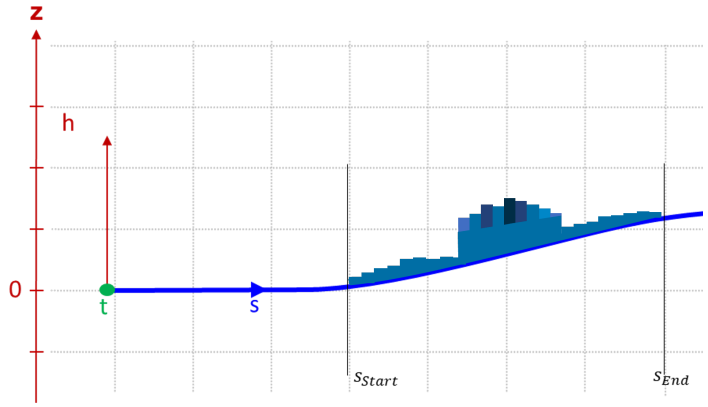
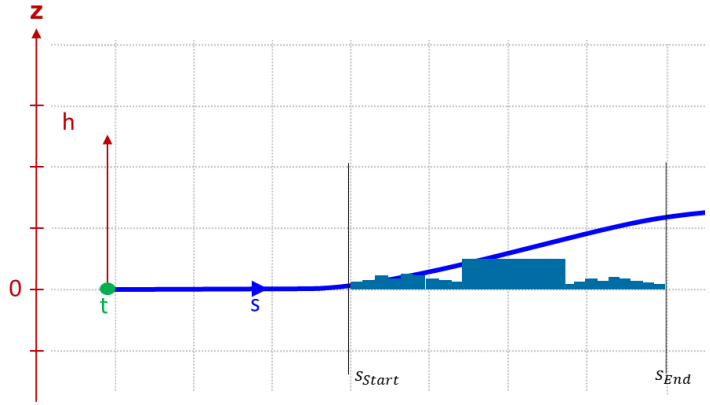
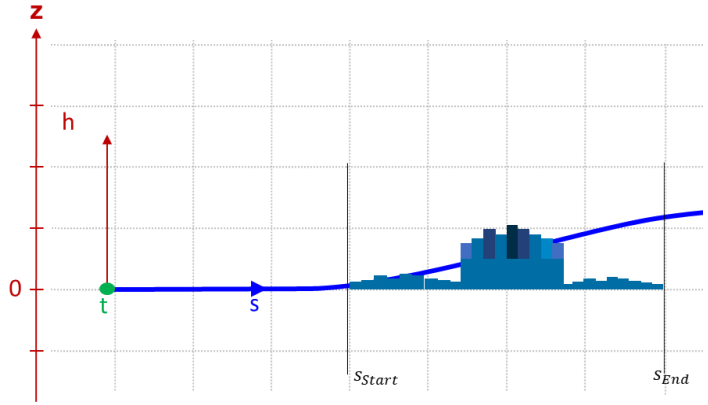
Thus, the surface height of an object is calculated as follows, using the `crgEvaluv2z` function from the ASAM OpenCRG C-API:

Processing math: 100%

$$\text{objectSurfaceHeight}(object, u, v) = \text{crgEvaluv2z}(crg, u, v) * z_{Scale}$$

The total height at the point of an object CRG depends on the attachment mode of the underlying road surface CRG (`t_road_surface_CRG`), provided there is one.

Table 107. Total height calculation by attachment mode

Road surface CRG attachment mode	hideRoadSurfaceCRG = true (default)	hideRoadSurfaceCRG = false
attached	OpenDRIVE height + object CRG 	OpenDRIVE height + road surface CRG + object CRG 
attached0, genuine, global	not allowed 	road surface CRG + object CRG 

Processing math: 100%

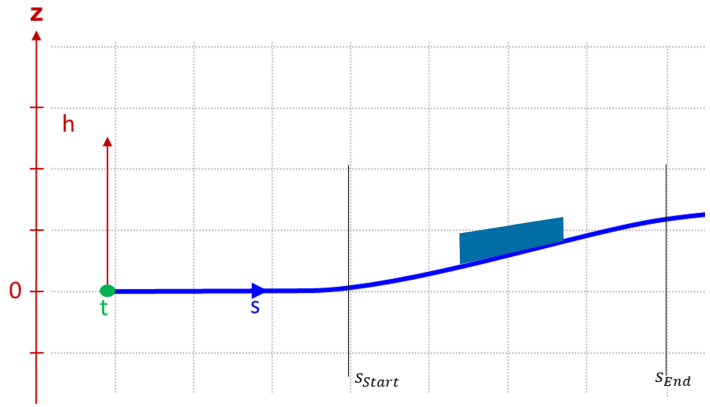
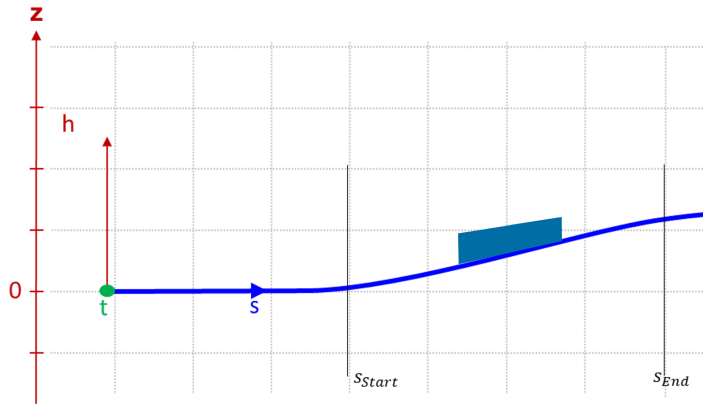
Road surface CRG attachment mode	hideRoadSurfaceCRG = true (default)	hideRoadSurfaceCRG = false
no road surface CRG	OpenDRIVE height + object CRG 	OpenDRIVE height + object CRG 

Table 108 summarizes the calculations for the different combinations. For a review of attachment modes, see Table 32, "Modes of connecting ASAM OpenCRG to ASAM OpenDRIVE" in Section 10.6, "Road CRG surface".

If `crgEvaluv2z` returns NaN, then the object has no defined height at that position. This is allowed in ASAM OpenCRG. In this case, the height of the road at that position shall be identical to the height defined in the rest of this standard, as if the object CRG were not present. The @hideRoadSurfaceCRG attribute has no effect at positions where `crgEvaluv2z` returns NaN. This enables the use of non-rectangular objects together with the @hideRoadSurfaceCRG attribute, for example, manhole covers.

Rules

The following rules apply to the use of CRG data in objects:

- asam.net:xodr:1.7.0:road.object.surface.only_for_angular_boxes: Only objects with angular bounding boxes may contain `<surface>` elements. Circular objects or objects with `<outlines>` elements shall not contain `<surface>` elements.
- Outside the bounding box, CRG data from the object shall be ignored.
- asam.net:xodr:1.7.0:road.object.surface.no_bounding_box_overlap: The bounding boxes of objects with `<surface>` elements shall not overlap.

Processing math: 100%

- asam.net:xodr:1.7.0:road.object.surface.identical_local_coordinates: The local coordinate system of the CRG shall be identical to the local coordinate system of the object to which it belongs. The reference line, inertial position, curvature, and heading of the CRG file shall be ignored.
- asam.net:xodr:1.7.0:road.object.surface.object_reference_on_overlap: An object with a `<surface>` element shall be referenced on all roads it overlaps, using `<object>` and `<objectReference>` elements.
- asam.net:xodr:1.7.0:road.object.surface.only_one_crg_file: An object shall not reference more than one CRG file.
- asam.net:xodr:1.7.0:road.object.surface.repeat_discretely_not_continously: Objects with `<surface>` elements may repeat discretely, but not continuously. See Section 13.2, "Repeating objects".
- To avoid skewed CRG surfaces, the @perpToRoad attribute should only be used for objects that are smaller than the local radius of the curvature of the road elevation.
- The @height and @zOffset attributes of an object with a `<surface>` element shall be ignored when calculating the surface elevation.
- asam.net:xodr:1.7.0:road.object.surface.crg_hidden_on_object_overlap: If a road surface CRG is present, that is, the CRG area overlaps the bounding box of the object and has any mode other than attached, then @hideRoadSurfaceCRG shall be false. True shall not be allowed.
- asam.net:xodr:1.7.0:road.object.surface.calculate_road_height: If `crgEvaluv2z` returns NaN, then the road height at that position shall be the ASAM OpenDRIVE height in addition to the road surface CRG, if it is present. The value of @hideRoadSurfaceCRG attribute shall have no influence.The value of @hideRoadSurfaceCRG attribute shall have no influence.

Related topics

- Section 10.6, "Road CRG surface"
- Section 12.12, "Junction CRG surface"

Processing math: 100%